Linear Regression: Prostate Cancer

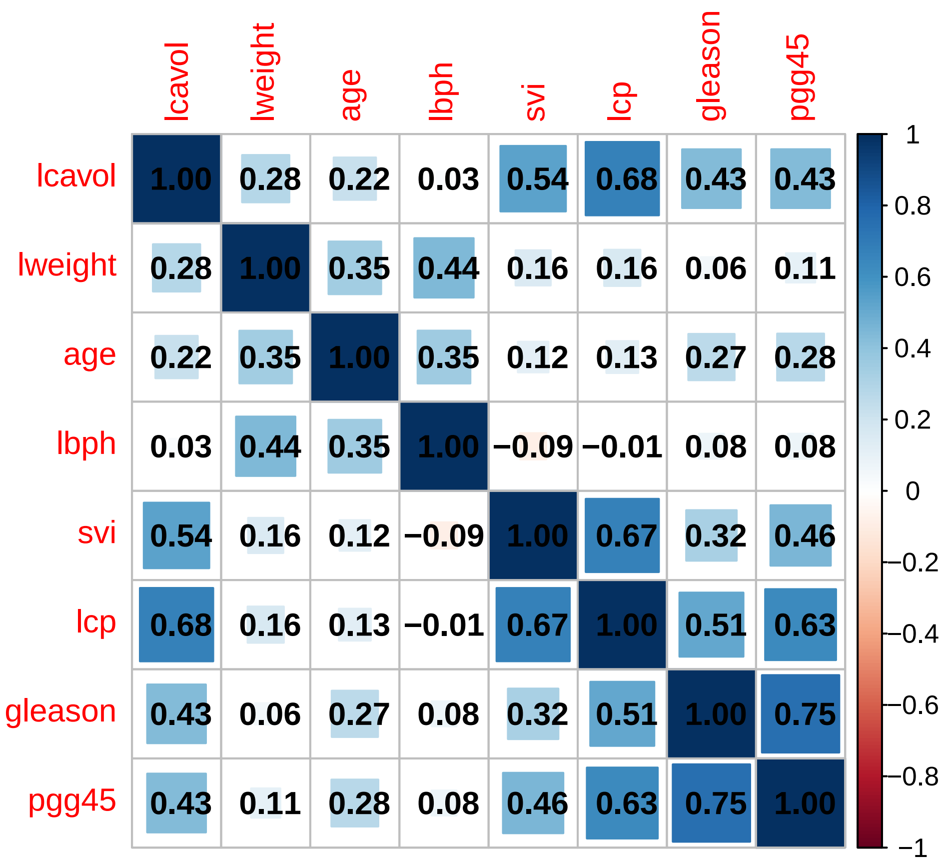
```r
rm(list = ls())
library(stargazer) ; library(stats) ; library(tidyverse) ; library(readr) ; library(corrplot) ; library
# Linear Regression : Prostate Cancer
# Data come from Stamey et al. (1989).
# Look a correlation between the level of prostate-specific antigen and
# a number of clinical measures in men who were about to receive a radical prostatectomy.
# The variables are :
# log cancer volume (lcavol)
# log prostate weight (lweight)
# age
# log of the amount of benign prostatic hyperplasia (lbph)
# seminal vesicle invasion (svi)
# log of capsular penetration (lcp),
# Gleason score (gleason)
# percent of Gleason scores 4 or 5 (pgg45).
# We first load the data and examine it.
data <- read_delim("prostate.data.txt", delim = "\t") # import the data

names(data)[1] <- "ID" # rename the ID column
head(data) ; summary(data) # visualize
```

```
## # A tibble: 6 x 11
##      ID lcavol         lweight   age lbph    svi lcp   gleason pgg45 lpsa  train
##   <dbl> <chr>            <dbl> <dbl> <chr> <dbl> <chr>   <dbl> <chr> <chr> <lgl>
## 1     1 "-0.579818495"    2.77    50 -1.3~     0 -1.3~       6 "  0" "-0.~ TRUE
## 2     2 "-0.994252273"    3.32    58 -1.3~     0 -1.3~       6 "  0" "-0.~ TRUE
## 3     3 "-0.510825624"    2.69    74 -1.3~     0 -1.3~       7 " 20" "-0.~ TRUE
## 4     4 "-1.203972804"    3.28    58 -1.3~     0 -1.3~       6 "  0" "-0.~ TRUE
## 5     5 " 0.751416089"    3.43    62 -1.3~     0 -1.3~       6 "  0" " 0.~ TRUE
## 6     6 "-1.049822124"    3.23    50 -1.3~     0 -1.3~       6 "  0" " 0.~ TRUE
```

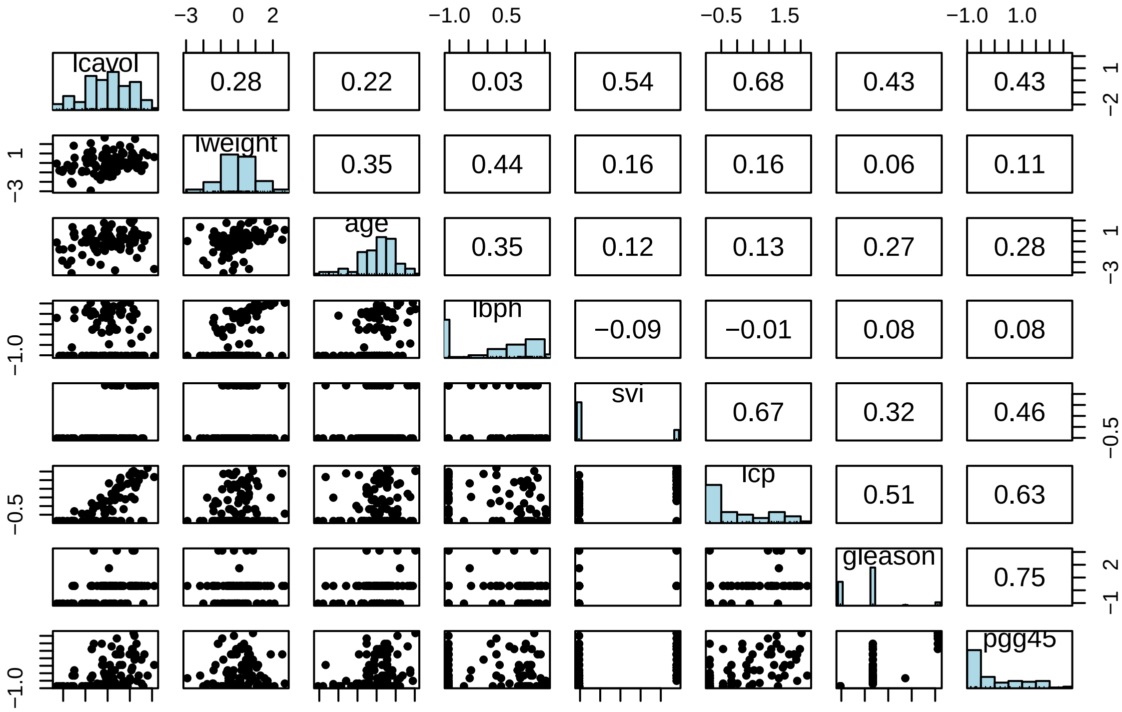
```
##        ID        lcavol             lweight           age
##  Min.   : 1   Length:97          Min.   :2.375   Min.   :41.00
##  1st Qu.:25   Class :character   1st Qu.:3.376   1st Qu.:60.00
##  Median :49   Mode  :character   Median :3.623   Median :65.00
##  Mean   :49                      Mean   :3.629   Mean   :63.87
##  3rd Qu.:73                      3rd Qu.:3.876   3rd Qu.:68.00
##  Max.   :97                      Max.   :4.780   Max.   :79.00
##      lbph                svi              lcp               gleason
##  Length:97          Min.   :0.0000   Length:97          Min.   :6.000
##  Class :character   1st Qu.:0.0000   Class :character   1st Qu.:6.000
##  Mode  :character   Median :0.0000   Mode  :character   Median :7.000
##                     Mean   :0.2165                      Mean   :6.753
##                     3rd Qu.:0.0000                      3rd Qu.:7.000
##                     Max.   :1.0000                      Max.   :9.000
##     pgg45               lpsa              train
##  Length:97          Length:97          Mode :logical
```

```
##  Class :character   Class :character   FALSE:30
##  Mode  :character   Mode  :character   TRUE :67
##
##
##
```

```r
vars <- c("lcavol", "lweight", "age", "lbph", "svi", "lcp", "gleason", "pgg45") # features
data <- data %>% mutate_at(append(vars, "lpsa"), as.numeric) # convert the the variables to numeric.
data <- data %>% mutate_at(vars, ~ scale(.) %>% as.vector) # standardize the features
corr_matrix <- cor(data[,vars]) # correlation matrix
corrplot(corr_matrix, method = "square", addCoef.col = 'black')  # associated plot
```




```r
# scatter plot matrices (SPLOM),
# with bivariate scatter plots below the diagonal,
# histograms on the diagonal,
#and the Pearson correlation above the diagonal.
pairs.panels(data[, vars], hist.col = "lightblue", density = F, smooth = F,
             ellipses = F)
```




```r
# We fit a linear model to the log of prostate-specific antigen, lpsa, after
# Randomly split the data into training set of size 67 and test set of size 30.
train_data <- subset(data, train == "TRUE") ; nrow(train_data)
```

```
## [1] 67
```

```r
formula <- reformulate(vars, response = "lpsa") ; formula
```

```
## lpsa ~ lcavol + lweight + age + lbph + svi + lcp + gleason +
##     pgg45
```

```r
model <- lm(formula, data = train_data)
summary(model)
```

```
##
## Call:
## lm(formula = formula, data = train_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.64870 -0.34147 -0.05424  0.44941  1.48675
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.46493    0.08931  27.598  < 2e-16 ***
## lcavol       0.67953    0.12663   5.366 1.47e-06 ***
## lweight      0.26305    0.09563   2.751  0.00792 **
## age         -0.14146    0.10134  -1.396  0.16806
## lbph         0.21015    0.10222   2.056  0.04431 *
## svi          0.30520    0.12360   2.469  0.01651 *
## lcp         -0.28849    0.15453  -1.867  0.06697 .
## gleason     -0.02131    0.14525  -0.147  0.88389
```

```
## pgg45        0.26696    0.15361   1.738  0.08755 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.7123 on 58 degrees of freedom
## Multiple R-squared:  0.6944, Adjusted R-squared:  0.6522
## F-statistic: 16.47 on 8 and 58 DF,  p-value: 2.042e-12
```

```r
# we compute the associated MSPE on the test data
test_data <- subset(data, train != "TRUE") ; nrow(test_data)
```

```
## [1] 30
```

```r
predictions <- predict(model, newdata = test_data)
MSPE_1 <- sum((predictions - test_data$lpsa)^2)/nrow(test_data) ; MSPE_1 # approx 0.52
```

```
## [1] 0.521274
```

```r
# In contrast, if we use the mean lpsa in the training set,
# we would get a MSPE of
MSPE_2 <- sum((mean(train_data$lpsa) - test_data$lpsa)^2) / nrow(test_data) ; MSPE_2 # 1.06
```

```
## [1] 1.056733
```

```r
MSPE_1/MSPE_2 - 1 # linear model reduces "base error rate" by about 50\%
```

```
## [1] -0.5067118
```